

Proposal for the 2026 NASA SUITS Challenge

Team **SIRIUS**

Spacesuit Immersive Reality Integrated Utility System



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Date: October 30, 2025

Technical Section.....	4
Abstract.....	4
Software and Hardware Design Description.....	4
User Interface.....	5
Web-Based Interface Core.....	5
Heads-up Display.....	6
Wrist Displays.....	7
Haptic 2-Axis Joystick.....	7
Full VR Mode.....	8
Networking.....	9
Concept of Operations (CONOPS).....	10
Rover Navigation.....	10
Egress.....	10
EV Navigation.....	11
LTV Repair.....	11
Ingress.....	11
Artificial Intelligence.....	11
Mission Assistant.....	11
System Summaries.....	12
Path Planning.....	12
Human-in-the-loop (HITL) Testing.....	13
Project Management.....	14
Technical References.....	15
Community and Industry Engagement Section.....	16
Community Outreach.....	16
Purpose of Outreach.....	16
Events for K-12th Students.....	17
Events for the Northeastern Community.....	17
Events for the Local Community.....	18
Media Outreach Plan.....	18
Industry Outreach.....	18
Administrative Section.....	20
Institutional Letter of Endorsement.....	20
Statement of Supervising Faculty.....	21
Statement of Rights of Use.....	22
Funding and Budget Statement.....	23
Budget Justification.....	24
Funding Sources.....	24
HoloLens 2 Loan Program.....	25
Appendix.....	26
Outreach Section.....	26
Appendix References.....	28

Technical Section

Abstract

SIRIUS (Spacesuit Immersive Reality Integrated Utility System) provides astronauts with digital capabilities designed to support lunar extravehicular activities (EVA). The system enhances astronauts' self-reliance on the lunar surface, reducing dependency on high-latency communication with mission control, and increasing mission efficiency and safety. Built on extended reality (XR) and artificial intelligence (AI), SIRIUS provides quick and easy access to information and system controls. Core features include a virtual workbench, an AI assistant, a haptic joystick, voice and gesture controls, and a customizable heads-up display (HUD).

After the SIRIUS team's successful field tests at the previous NASA SUITS challenge, our system has gone through rigorous improvements to redesign the user experience and organize system development. The hardware stack has been redesigned to offload main tasks from the Quest 3 headset to a dedicated desktop for processing-intensive operations. The new architecture features a web-based networking layout that is simpler and more resilient than our previous iteration. These improvements result in a system that is more stable and flexible, providing a smoother and cleaner experience for the astronaut.

Software and Hardware Design Description

All network coordination, AI models, image recognition, and path solving in SIRIUS are performed on a small form-factor custom-built micro-ITX desktop. Equipped with an Nvidia graphics processing unit (GPU), it enables us to solve parallelizable tasks efficiently and effectively.

Our desktop acts as the central hub for our networking and communication with NASA's Telemetry Stream Server (TSS). It interprets information from the TSS and sends it out to the individual crew members' head-mounted displays (HMDs) and wrist displays. When a crew member makes an input on any of their devices, it is sent back to the desktop which then forwards it to the TSS. This enables our desktop to be the source of truth for our system. If an external device goes down or loses connection, our desktop will still maintain connectivity with the TSS, ensuring system redundancy. To support this architecture and maintain consistent visuals across devices, we will host a local webpage on the desktop system. This webpage provides visuals and controls for all major parts of our system and can be accessed across all devices.

To interact with this webpage, our crew members wear an HMD and wrist-mounted displays to visualize and interact with system information. The crew member inside the pressurized rover (PR) wears a Meta Quest 3 HMD. The astronaut outside the rover, engaging in the extravehicular activity (EVA) components of the lunar EVA scenario, wears an XReal Pro One HMD. Both headsets provide pass-through augmented reality, allowing the user to see the real world with overlaid virtual elements. Additionally, the system supports multiple microcontroller-based peripheral interfaces, including touch-screen displays running on Raspberry Pi boards and a haptic joystick powered by an ESP32.

User Interface

Web-Based Interface Core

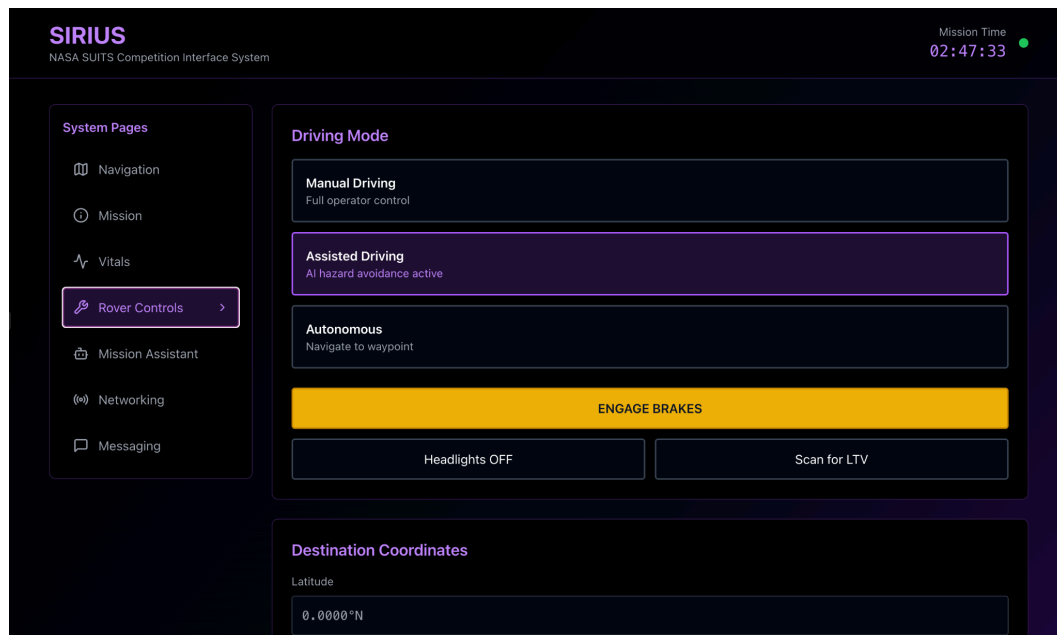


Figure 1: SIRIUS web-based interface wire frame.

SIRIUS's core user interface (UI) is built around a dynamic webpage-based system hosted on our central computer. Designed in JavaScript's React web framework, this interface offers sleek interactions and streamlined displays for all system and mission information. System interactions and interfaces are organized through a library of web-based widgets allowing for effective and modular communication across devices.

Widgets include system vitals, a messaging hub, a mission task checklist, a mission clock, a compass, a map of the lunar surface, and more. Using these widgets, SIRIUS can populate our system information and controls across various devices, maintaining consistency in visuals and controls while providing a highly modular and customizable UI.

Additionally, the use of cascading style sheets (CSS) allow the user to customize the colors, text, and fonts of the system to accommodate various lighting and accessibility needs.

The entire webpage is hosted locally on our system hub. This allows the system to be highly redundant; if at any point devices in our system fail or lose connection, all controls and system information will remain accessible directly through the hub. Shown above in Figure 1 is the basic layout of the backup UI hosted on the system hub.

Heads-up Display

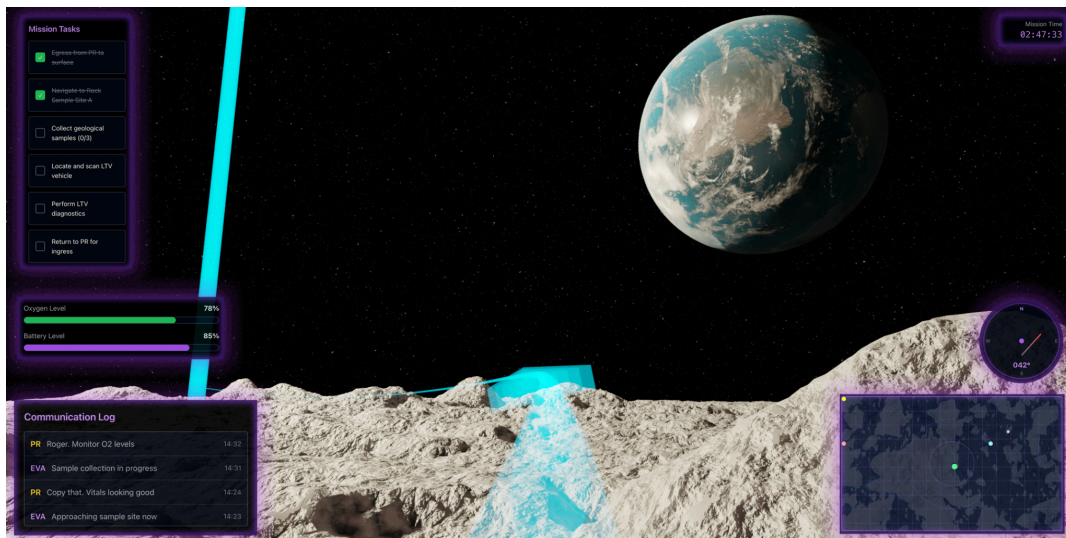


Figure 2: EVA member HUD wireframe.

The HUD (Figure 2) provides the crew members with a constant, accessible, and ergonomic display of system information. System widgets display in the peripherals of the user's vision, allowing for easy access to vital information without inhibiting current tasks. The default layout of the HUD consists of navigation and mission info widgets supplied from SIRIUS's webpage. These elements can be swapped and rearranged for full customization, and in the event that the display ever impedes the user's vision, the HUD may be hidden with the quick voice command "HUD off."

Additionally, the HUD maintains toggleable views for navigation and hazard overlays. The navigation overlay displays the optimal path to the user's next waypoint destination as well as a breadcrumb trail, ensuring that they do not get lost in an unfamiliar landscape. SIRIUS also utilizes image processing techniques to highlight environmental hazards in front of the user, promoting safety in harshly-lit environments like the lunar south pole.

Wrist Displays



Figure 3: Left wrist display.



Figure 4: Right wrist display.

Running on two individual Raspberry Pi microcomputers are touch-screen wrist-mounted displays providing both the PR and the EVA crew members with quick access to vital information and commonly used system widgets. The default wrist display layout populates the crew member's left wrist with figures relating to the user's vitals and resources (Figure 3). The right wrist displays navigation widgets intended for use during EVA (Figure 4). These wrist displays offer the user the ability to visualize system widgets outside of their direct field of view, while still being quickly accessible.

Beyond viewing information, these touch-screen displays additionally offer the user a method for precise data entry and interaction. When interacting with portions of the system that require inputting digits or text, the display will automatically switch to a touch-screen keyboard on the user's non-dominant wrist.

Haptic 2-Axis Joystick

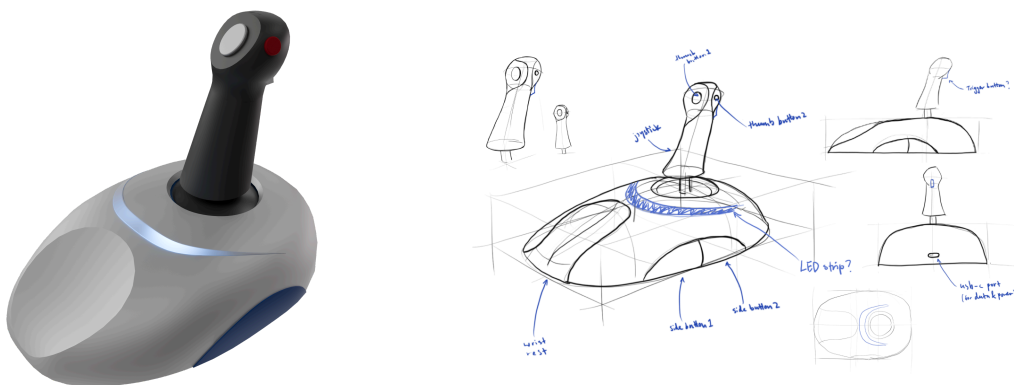


Figure 5: Haptic joystick design plans.

To control the PR, SIRIUS is equipped with a haptic 2-axis joystick (Figure 5). The joystick is equipped with an ESP32 microcontroller and dual serial bus servos for haptic feedback. This setup allows us to create a joystick that directly interfaces with our

autopilot system, allowing for both manual and autonomous navigation. When the autopilot controls the joystick, the stick physically actuates, allowing the operator to feel the autopilot's intentions. If the operator disagrees with the autopilot, they may provide physical input to override the autopilot's control. Additional button inputs are integrated into the top of the stick. Buttons integrated into the joystick handle allow for "eyes-off" control of critical secondary functions. These are mapped to tasks such as dropping points of interest, activating the lunar terrain vehicle (LTV) "wake-up call" signal, and cycling through operation modes. In addition to force feedback, the haptic actuators generate vibrations for alerts and to provide distinct warnings for hazard avoidance. Additionally, these actuators enable our "warmer/colder" feature to pulse as the rover approaches the LTV's position.

The joystick has three driving modes with different levels of driving assistance. Our first mode is manual navigation. In manual mode, the joystick will not provide any active force feedback. The servos fulfill the role of springs in a traditional, non-haptic joystick, recentering the joystick to the origin. Our second mode is hybrid navigation. In hybrid mode, the joystick will provide limited force feedback to the user based on the autopilot/automation system. This feedback may be used to lightly steer the pilot towards the optimal route, or away from a detected obstacle. Finally, we have full autonomous mode. In full autonomous mode, the rover will move fully hands-free with no input from the crew member. In this mode, the joystick motors will move to correspond to the inputs of the autopilot, visualizing the autonomous decisions and allowing the astronaut to regain control if needed.

Full VR Mode

An advantage of building SIRIUS in Unreal Engine 5 is the ability to integrate NASA's DUST environment into our HMD. This allows us to offer a full virtual reality (VR) mode toggle while driving the PR. Toggling full VR mode shows the driver of the PR the full lunar surface around the vehicle, enabling a more detailed view of the surrounding terrain to increase mission safety and effectiveness. Additionally, the use of full VR mode makes it much easier to spot the LTV from the PR during the search phase of the mission. In a real-world scenario, a 360-degree camera could be incorporated above the PR to enable this technology.

Networking

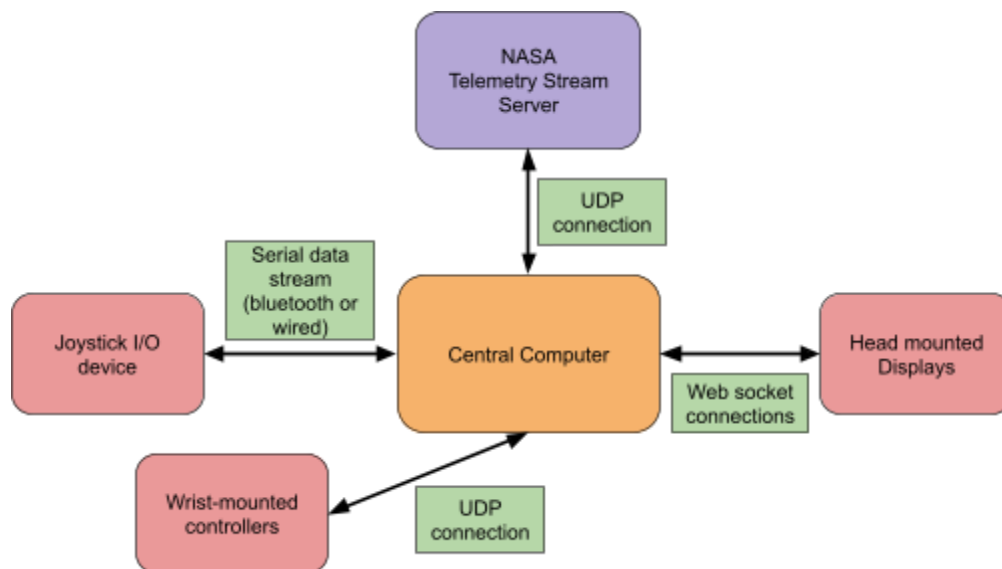


Figure 6: SIRIUS network server architecture.

SIRIUS employs a distributed network architecture (Figure 6) centered around the micro-ITX desktop hub that serves as the system's central communication node. This hub manages all data flow between NASA's Telemetry Stream (TSS), crew devices, and autonomous systems, ensuring reliable and low-latency information exchange across the lunar EVA mission.

The desktop hub acts as the primary interface with the TSS, receiving data and transmitting commands back to mission control. All crew devices including HMDs, wrist displays, and the haptic joystick communicate through this central hub rather than directly with each other or the TSS. This star topology provides a single source of truth for system state, simplifying fault isolation and recovery. This represents a key improvement from last year's system where all networking was processed onboard our Quest 3 headset, a design that proved inflexible and difficult to troubleshoot during test week.

Our new system utilizes multiple communication protocols optimized for different device requirements. The locally hosted webpage interface is delivered over Wi-Fi to the Meta Quest 3 and XReal Pro One HMDs, as well as to the Raspberry Pi-powered wrist displays. This web-based approach ensures consistent rendering across platforms while minimizing development overhead. The haptic joystick communicates with the hub via a Bluetooth serial connection through its ESP32 microcontroller, providing low-latency transmission of position data and receiving motor control commands for force feedback.

Concept of Operations (CONOPS)

Rover Navigation

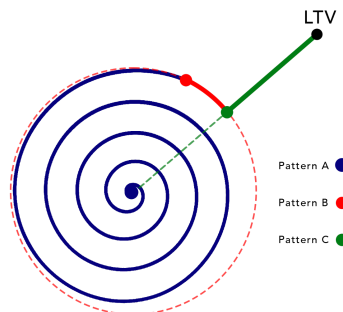


Figure 7: Rover search patterns.

At the beginning of the mission, the rover will navigate to the last broadcast position of the LTV. Upon arrival, the rover will begin broadcasting the LTV wake-up signal and start our search protocol (Figure 7). The pressurized rover will then navigate in an outward spiral search pattern (pattern A), using the last known position of the LTV as an origin point. Once the wake-up signal receives a response from the LTV, we will switch patterns to a perfect circle around our search origin point (pattern B). During this pattern, we will monitor our wake-up signal for the point where the "warmer/cooler" feature reads "cooler", meaning that we are moving farther away from the LTV. At this point, we will mark a waypoint on our map and, drawing a line between this point and our search's origin point, we will find a straight linear trajectory towards the LTV's position. We then follow this line until we establish visual contact with the LTV (pattern C). All rover navigation input will be made using our haptic joystick.

Egress

During egress, all necessary steps to safely depart the cabin will populate on the EV member's task list, ensuring that proper safety protocols are followed. If at any point the EV member breaks protocol and deviates from the proper egress procedure, an audible system warning will occur notifying them of their mistake. Throughout this process, the PR crew member will monitor the EV member's progression through system notifications to help ensure their safety.

EV Navigation

After leaving the PR, the EV navigation will begin on the lunar surface. The EV member will have a detailed HUD (Figure 2) composed of widgets and overlays to assist with navigation. This HUD is outlined in detail in the User Interface section.

LTV Repair

Once the EV member arrives at the LTV, they will go through the necessary steps for repair. To begin, the crew member will notify the AI assistant that they have arrived at the LTV, triggering the assistant to populate the task list widget with the necessary exit recovery mode (ERM) procedures. After completing ERM tasks, the AI assistant will enter deep think mode to diagnose the problems with the LTV. Necessary physical repairs will be highlighted in the EV members' vision by the AI assistant.

Ingress

After successfully repairing the LTV, the EV member will return to the PR. They will receive the PR's location from the waypoint manager and, after setting it as a new destination waypoint, navigation widgets, and the navigation overlay will update to display a safe path back to the airlock.

Artificial Intelligence

Mission Assistant

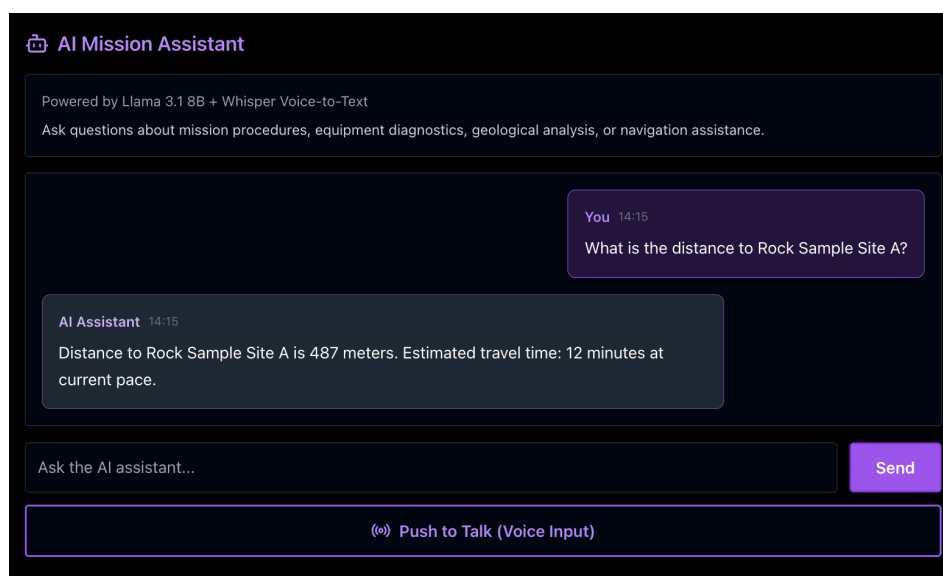


Figure 8: Chat UI for Mission Assistant in SIRIUS server.

Available to astronauts at all times is the AI-powered Mission Assistant accessible through both voice commands and a comprehensive chat user interface integrated within the SIRIUS server. Our Mission Assistant employs a retrieval-augmented generation (RAG) architecture, grounding all responses in mission-specific documentation for SIRIUS systems, SUITS operational procedures, and real-time telemetry data. This approach ensures that the assistant provides accurate, contextually relevant information while

minimizing hallucinations and maintaining the focus on simplicity and robustness required for mission-critical operations. The chat interface (Figure 8) provides a persistent conversational history. This allows astronauts to reference previous guidance and maintain context across complex multi-step procedures, while the voice interface offers hands-free access during physically demanding tasks.

The SIRIUS server hosts a model context protocol (MCP) that provides the Mission Assistant with direct access to all rover and astronaut telemetry streams. Through this integration, the assistant can plan waypoints using current navigation data, walk astronauts through mission procedures with awareness of completed steps, alert crew members to emerging hazards based on continuous monitoring of vital signs and environmental conditions, and propose rover commands that require explicit astronaut approval before execution. For voice interaction, we run instances of Llama 3.1 8B for natural language processing and OpenAI's Whisper for voice-to-text conversion, chosen for being open source, allowing local inference on the SIRIUS desktop. The wake word "SIRIUS" activates voice input, which is then processed alongside the RAG-retrieved context and real-time telemetry data. All system statistics mentioned by the assistant are replaced with absolute values directly from the TSS, further ensuring accuracy and reliability in mission-critical scenarios.

System Summaries

Beyond our lightweight voice assistant, we also include a separate model for deep thinking to offer crew members detailed information for troubleshooting unexpected problems and issues that may arise on the mission. This model uses Google Colab's Notebook LLM with mission documentation to assist with complex problem-solving while minimizing hallucinations. This model is accessible through both the standard voice assistant with the command "SIRIUS analyze" and through relevant system widgets. Widgets that display system information will include an "analyze" button. When clicked, this button displays a floating screen with a summary generated by the model.

Path Planning

For both PR and EVA navigation, we have built upon our perception and planning software stack from last year, making it more modular and extensible. Through our new haptic joystick, SIRIUS supports both simple driver assistance with force-feedback and complete supervised autonomous automation, depending on the astronaut's preferences.

Our autonomous stack aggregates provided LIDAR data into a local elevation map, leveraging the localization (latitude, longitude, orientation) from the DUST environment. This local elevation map displays a high-resolution view of the area immediately around the rover. This can then be downsampled into a lower-resolution global elevation map that

can be utilized throughout the duration of the mission to recall previously visited locations. Additionally, the local elevation map is fused into a local cost map, approximating the risk of terrain around the rover. Vetted at the 2025 SUITS challenge, we utilize Model Predictive Path Integral (MPPI) control, a state of the art algorithm designed for robust path planning in rough and unpredictable terrain. MPPI models the kinematics of the rover and simulates trajectories over the cost map to direct the rover to a desired location while minimizing risk (Williams et al., 2016).

To better apply GPU-based hardware acceleration of MPPI, our autonomous stack is written in Python leveraging Nvidia's CUDA platform running on the SIRIUS desktop, interfacing directly with the TSS and DUST. Autonomous control generated is output to the haptic joystick to provide feedback to the driver, or to move the joystick autonomously if the driver is hands-off.

Human-in-the-loop (HITL) Testing

As a part of SIRIUS's development, extensive human-in-the-loop testing will be conducted to ensure that our system maintains high standards of usability and functionality. By mid-November, we will have full prototypes of our web-based widgets and our haptic joystick, allowing for initial testing and feedback on our designs. Members of the SIRIUS team will test these aspects of our systems at our weekly meetings, providing feedback to iterate and refine our design. Additionally, we will hold public testing days every two weeks to collect external feedback from non-teammembers.

By mid-January, we will have a refined version of our website widgets and our haptic joystick, as well as basic hand gestures implemented in our HMD for input. Using these systems, we will partner with Northeastern's Spatial Intelligence Research Group (SINRG) to conduct an extensive user-based study for our system. This research study will be centered around exploring the effectiveness of multi-tasking while driving on the lunar surface using our haptic joystick-assisted driving modes (manual, autonomous, and hybrid navigation) in parallel with a gesture-based UI. Our hybrid driving is designed to allow the astronaut to shift their context windows effectively between scientific tasks and safe navigation, both vital to completing a successful lunar mission. This research study will be conducted using standardized user study techniques and include simulated tasks to measure user performance alongside a standardized questionnaire.

Continuing development, we will have a version of our system with all implemented devices by mid-March. At this point, we will begin the final phase of our system testing: mission run-throughs. We will run four mission run-throughs during March and April. These run-throughs will take place outside, in a well-lit area of a turf field following all proper safety protocols. Run-throughs will simulate all processes of the SUITS EVA scenario as closely and accurately as possible to find bugs or missing features in our

system. Additionally, as part of these run-throughs, we will stress-test our communications stack and measure connectivity across our devices and networks.

All questionnaires in our user tests will be based on the NASA Task Load Index (NASA-TLX) assessment tool (Hart & Staveland, 1988) to maintain consistent and reliable feedback. Since our system incorporates full VR elements, we will include an evaluation for VR-induced nausea in our system using the Virtual Reality Neuroscience Questionnaire created by Kourtesis et al. (2019). To maintain hygiene and safety, all demos and tests will be performed with disposable covers on HMD devices, and all hardware and equipment will be thoroughly wiped and cleaned between users.

Project Management

Northeastern SUITS 2026 - Gantt Chart

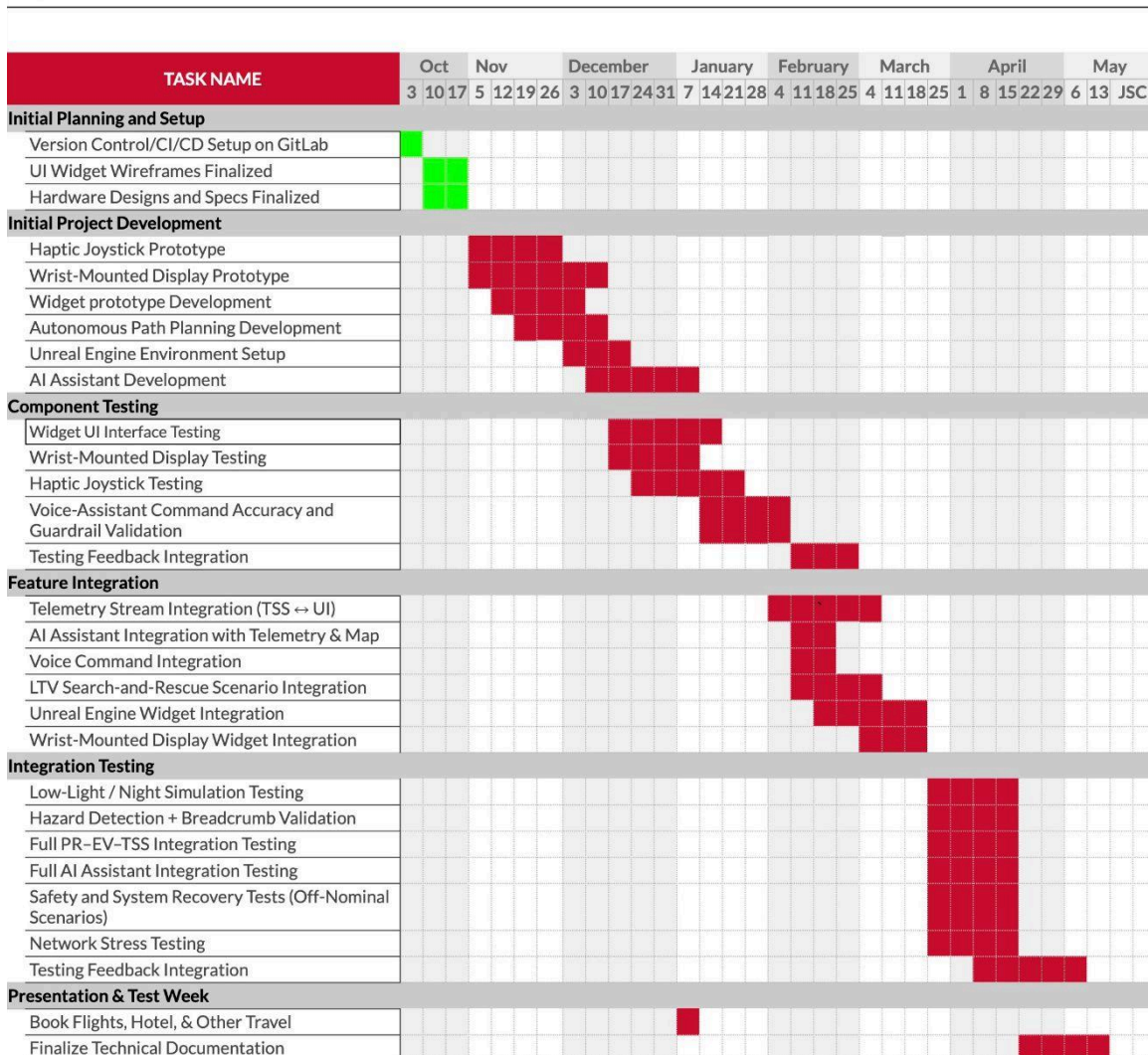


Figure 9: Development Gantt chart.

Team SIRIUS's project development will follow an agile development approach. Tasks will be split into smaller two-week tickets that we will then assign to team members. These tickets will be assigned and managed through our Notion home page.

Figure 9 lays out SIRIUS's expected development timeline. Our initial planning and setup phase has already been completed, allowing us to begin production on our system prototypes. We will be developing prototypes until early January, with initial component testing in mid-December. Feature integration will take place from February to late March, allowing for a month of integration testing and mission run-throughs. This development structure allows for detailed systems and HITL testing, while allotting time to integrate feedback and fix unexpected bugs and technical issues.

Technical References

- Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In P. A. Hancock & N. Meshkati (Eds.), *Human mental workload* (pp. 139–183). North-Holland.
[https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)
- Kourtesis, P., Collina, S., Doumas, L. A. A., & MacPherson, S. E. (2019, November 26). Validation of the Virtual Reality Neuroscience Questionnaire: Maximum Duration of Immersive Virtual Reality Sessions Without the Presence of Pertinent Adverse Symptomatology. *Frontiers in human neuroscience*, 13, 417.
<https://doi.org/10.3389/fnhum.2019.00417>
- Williams, G., Drews, P., Goldfain, B., Rehg, J. M., & Theodorou, E. A. (2016). Aggressive driving with model predictive path integral control. *2016 IEEE International Conference on Robotics and Automation (ICRA)*, 1433–1440.
<https://doi.org/10.1109/icra.2016.7487277>

Community and Industry Engagement Section

Community Outreach

Event	Date	Type of Event	Target Audience
Northeastern Splash	November 8th	Educational activity	8th-12th students
Building Bridges	December 5th	Educational activity	9th-12th students
Museum of Science Computer Science Weekend	December 13th-14th	Tabling opportunity	Boston community
Engineering for Everyone Expo	February 20th	Tabling opportunity	K-8th students
Boston Public Schools STEM Fair	March 7th	Tabling opportunity	6th-12th students
STEMout	Winter/Spring 2026	Educational activity	K-8th students
Northeastern Space & Technology Expo	Spring 2026	Speaker event, tabling opportunity	Undergraduate and graduate students

Table 1: Planned community outreach events schedule.

Purpose of Outreach

By providing outreach to students and community members in the local Boston area, we will spread awareness and excitement about STEM and lunar exploration. Our outreach plan is specifically designed to introduce them to NASA's Artemis program, engage them with STEM-based learning activities, and encourage them to continue fostering their interest in STEM, extended reality (XR) technologies, and space exploration. Through partnerships within Northeastern University and with various Boston organizations, we will participate in a variety of outreach initiatives to achieve our goal of positively engaging with the communities we are lucky to be a part of.

Events for K-12th Students

On November 8th, we are participating in Northeastern's Fall Splash. Splash is a free educational event open to local middle and high schoolers. At Splash, undergraduates at Northeastern University collaborate to teach engaging and informative classes. We will be teaching an hour-long interactive class & workshop that introduces students to virtual and augmented reality (appendix Figure A1).

On December 5th, we plan to collaborate with Northeastern's Center for STEM at Building Bridges. This day-long event allows high school students to explore the College of Engineering at Northeastern University, participate in various interactive activities related to the engineering focuses offered at Northeastern, and introduce them to potential future educational and career opportunities involved in engineering. We intend to give a presentation to these students that provides them with an overview of NASA's Artemis programs and SUITS, as well as showcase demonstrations of SIRIUS and our artificial intelligence system. We will also answer any questions students may have about these technologies and encourage them to continue exploring engineering as a field of study.

On February 20th, we will be participating in Northeastern's Engineering for Everyone Expo. This event invites K-8 students and their families to Northeastern University to participate in interactive lessons that focus on engineering and make connections with several students currently pursuing engineering at Northeastern University. At the Expo, we will introduce students and their families to XR to incite interest in the technology.

On March 7th, we plan to run an in-person table at the 2026 Boston Public Schools STEM fair. At this expo, we will be showcasing our SUITS system to over a hundred local elementary and middle public school students to get them excited about space exploration and XR technology.

Working together with STEMout, a Northeastern University student organization designed to make STEM more accessible and engaging for Boston students, we will create and run STEM activities and lessons for K-8 students at the Jamaica Plain Branch of the Boston Public Library. We will also run after-school STEM activities for students at the Curley K-8 School. Lessons will be modeled around an Hour of Code structure.

Events for the Northeastern Community

To get our school community interested, engaged, and involved with XR technology and its applications (particularly in space exploration), we intend to present at a variety of Northeastern University events. The first event we will attend is the February College of Computer Science's Mixed Reality Expo. At this event, we will be surrounded by peers who share our interest in XR technologies and their potential. We plan to partner with

Northeastern's Virtual Reality club to showcase current events in the field of XR and encourage the development of XR research conducted by students and faculty within our community (appendix Figure A2). We will also be highlighting the NASA SUITS mission and how our (and our partners') system functions.

The next event we plan to participate in is Northeastern's annual Space and Technology Expo. This event is held for Northeastern students to inform them of space-related opportunities through the school. At this event, we will demonstrate everything we have learned and accomplished so far participating in SUITS, as well as garnering interest among the student body for participation in next year's SUITS challenge.

Events for the Local Community

The Students for the Exploration and Development of Space and Team SIRIUS have been invited to table at the Boston Museum of Science's Computer Science weekend on December 13th and 14th (appendix Figure A3). Through this event, we have the opportunity to demonstrate our system to the Boston community and to engage families and younger children with space exploration and the Artemis missions. For older children and adults, we will demonstrate the SIRIUS system while collecting informal feedback. For younger children, we will provide a "design your heads-up display" coloring activity (appendix Figure A4).

Media Outreach Plan

Media outreach will be run through our student-run social media accounts. Here we will post progress on the SUITS challenge and inform others about any developments in XR technologies, NASA's programs, and SIRIUS. We will also continue to partner with Northeastern-run and student-run publications to help keep us engaged and active within our University's community. These publications include the Huntington News, NUSci, Northeastern's STEM blog, and news@Northeastern. Through these publications, we will publish articles on NASA SUITS and the progress we've made with this project.

Industry Outreach

For the first time this year, we will be partnering with Northeastern University's SINRG labs to conduct a research study based on the SIRIUS system. This research study will use our haptic joystick and XR interactions. We will be seeking answers to the following research questions:

To what extent does human-involved autonomous driving on the lunar surface allow astronauts to shift their attention from navigation toward in-depth scientific analysis?

Which interaction techniques most effectively support decision-making in XR simulations, enabling astronauts to explore both overview and detailed terrain data while managing multiple concurrent mission activities?

This research study will thoroughly test the practicality and usability of SIRIUS's unique system while simultaneously contributing important research to the scientific community. We will write a detailed research paper and submit it to IEEE ISMAR 2026. Additionally, we will write an abstract based on our initial system and apply to demonstrate our prototype at the conference. Completing this study will help contribute to research around using novel technologies in lunar space exploration and provide SIRIUS team members with valuable industry experience.

Administrative Section

Institutional Letter of Endorsement



Northeastern University
College of Engineering

October 28, 2025

NASA SUITS Competition Committee
NASA Johnson Space Center
2101 NASA Parkway
Houston, TX 77058

Office of the Dean
230 Snell Engineering
Center
260 Huntington
Avenue
Boston, MA 02115
617.373.5847
fax: 617.373.8504
northeastern.edu/coe

Dear NASA SUITS Committee,

I am writing to confirm that Northeastern University fully supports the participation of our student organization, Students for Exploration and Development of Space (SEDS), in the 2026 NASA SUITS competition. We are aware of their efforts to develop an augmented reality heads-up display for NASA spacesuits as part of the challenge, and we endorse their involvement in this initiative. Should you require further information, please feel free to contact me.

Sincerely,

A handwritten signature in black ink, reading "Gregory D. Abowd".

Gregory D. Abowd
Dean, College of Engineering

Statement of Supervising Faculty

As the faculty advisor for an experiment entitled SIRIUS proposed by a team of higher education students from Northeastern University, I concur with the concepts and methods by which the students plan to conduct this project. I will ensure the student team members complete all project requirements and meet deadlines in a timely manner. I understand any default by this team concerning any project requirements (including submission of final report materials) could adversely affect selection opportunities of future teams from their institution.

Faculty Advisor Signature: *Mallesham*

Mallesham Dasari

October 30 2025

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Faculty Advisor Signature: *Malles*

Team Signatures:

Joshua Goldman

Nico Rising

Sania Hasan

Catherine Atzelby

James Chan

Lawrence Xie

Thomas Yi

Aidan Hansen

Jason Jathanna

Emily Don

Tony Sojan

Tianhong Zhong

Page Patterson

Brandon Petersen

Milo Lin

Funding and Budget Statement

Item	Cost
Flights (Boston to Houston)	\$2,560
Hotel Costs	\$4,200
Ground Transportation	\$1,000
Operations	\$200
Hardware	\$2,000
Miscellaneous	\$200
Total	\$8,100

Table 2: Budget overview.

Budget Justification

Outlined in Table 2, our budget of \$8,100 covers a comprehensive list of all expected expenses from participating in the NASA SUITS competition. We have budgeted \$2,800 for flights based on a round-trip estimate from Boston to Houston at \$320 per person and an additional buffer to account for fluctuations in prices and changing airfare. Our hotel costs cover a seven-night stay in Houston from May 16-22 (based on the scheduling of last year's challenge). Two members will stay in each room, requiring a total of four rooms. At an estimated rate of \$150 per room per night, this totals \$4,200.

Additionally, we have allocated \$1,000 for ground transportation. This allotment covers the rental fees for obtaining two sedan-sized vehicles for local transportation between the hotel, competition venue, and any off-site activities during test week. This provides our team with the necessary flexibility for commuting in Houston. We have also budgeted \$200 for operations, covering general expenses such as materials, printing, headset sanitation wipes, and stickers needed for outreach and community events.

Our hardware costs include purchasing a pair of XReal One Pro glasses, which cost \$650, and building a micro ITX desktop for \$850. We will be using previously acquired Meta Quest 3 headsets from participating in the challenge last year. Additionally, we will purchase hardware for the haptic joystick and wrist-mounted touch-screen controllers. This hardware includes an ESP32 microcontroller (\$15), two serial bus servos (\$45), two Raspberry Pi 5 microcomputers (\$120), two touch-screen displays (\$120), and three

battery packs (\$50). Additionally, we have budgeted an extra \$150 for wires, breadboards, 3D printer filament, and any unforeseen additional costs.

Lastly, a miscellaneous fund of \$200 is allocated to cover general unforeseen expenses. This fund will cover any unexpected costs associated with outreach events, public demonstrations, workshops, and booking meeting spaces. By allocating this additional fund, we ensure that our team has flexibility with our funds.

This budget comprehensively covers all costs associated with participating in the NASA SUITS challenge, while allowing us to have flexibility to adapt to any unanticipated costs or challenges.

Funding Sources

As a project within Northeastern University's Students for the Exploration and Development of Space organization and having successfully participated in SUITS last year, we have received a \$4,480 budget that will cover hardware, software, operations, ground transportation, and miscellaneous expenses. Additionally, Northeastern's College of Engineering has a separate fund to support outreach events. We can use this fund to help cover any unexpected costs relating to our planned outreach. Northeastern University's College of Engineering and College of Sciences have grants in place to provide funding for students' flights associated with participating in challenges like NASA SUITS. Last year, our flights were entirely reimbursed by the school, and we will follow the same process this year to ensure that they are covered.

HoloLens 2 Loan Program

We do not require a loaned device since we already have access to Quest 3 devices and have the funds necessary to purchase all other needed hardware.

Appendix

Outreach Section

E701: Introduction to Virtual Reality!

Difficulty: *

Teachers: [Joshua Goldman](#)

Interested in Virtual Reality? Come to our class to learn about, and experience, VR technology!

The class will be taught by Northeastern University's NASA SUITS team, a group that competes in NASAs annual Virtual Reality development challenge to create VR systems to help assist astronauts.Come and learn about VR, play games, and try out our system!

Class Style
Activity

Meeting Time
Section 1: Sat, Nov 08, 11:50am--12:50pm

Grades
8 - 12
Enrollment
Section 1: 4 (max 25)

**NORTHEASTERN PROGRAM FOR
TEACHING BY UNDERGRADUATES**

Figure A1: Screenshot of our registered Splash outreach class

**NUVR CLUB Northeastern** <nuvr.neu@gmail.com>
to me ▾

Hi Joshua Goldman,

I hope you are doing well.

Thank you for telling us about the SUITS project you are working on. Please keep me updated on outreach events and how we can work together to bring interesting VR projects to the Northeastern Community.

Have a wonderful weekend!

Best,
NUVR Club

Figure A2: Screenshot of talks for an outreach event with Northeastern University Virtual Reality.

CSEd Weekend 2025 Sign-up



Olivia Perdew-Kenny
to me ▾

Wed, Oct 8, 10:43 AM ☆ 😊 ↩ ⋮

Good Morning,

I wanted to follow-up on my previous email to share more about our plans for this year's Computer Science Education Weekend.

As you know, the event is set to take place on Saturday, December 13th and Sunday, December 14th and will run from 10am-4pm each day. Community group activities and demonstrations are scheduled to be open from 11am-3pm. We do ask, however, that all participating groups commit to being onsite from 10:30am to 3:15pm to allow time for set-up and clean-up.

If your team can attend again this year, please complete our [Activity/Demo sign-up form](#) to share your availability and additional scheduling details by Friday, November 7th so that we can include you in our activity passport and other print materials.

To show our appreciation for your help running this exciting event, all community partners will once again receive free parking at the Museum of Science, lunch, and a CSEd Weekend t-shirt!

Please let me know if you have any questions or concerns.

Thank you!
Olivia

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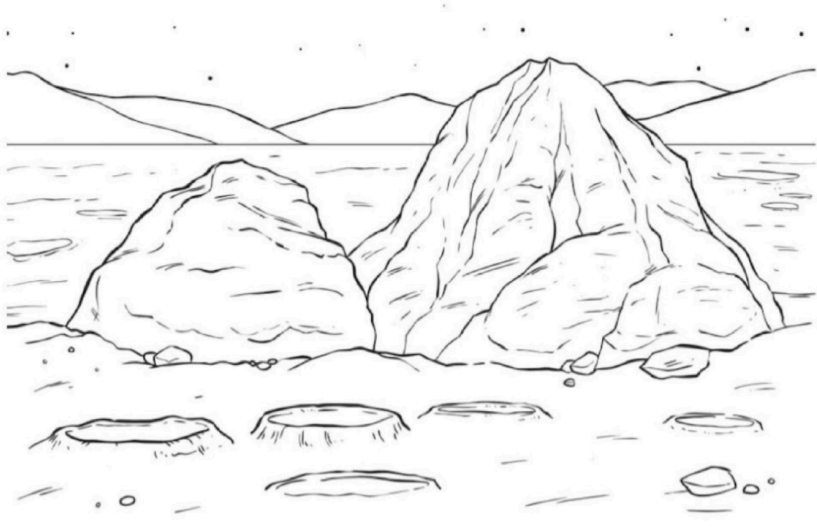
Olivia Perdew-Kenny | *she/her/hers*
Youth Engagement Program Manager | Tuesday - Saturday
Museum of Science | One Science Park | Boston, MA 02114
617-589-0338 | operdewkenny@mos.org

Figure A3: Screenshot of our invitation to the Boston Museum of Science for Computer Science Weekend Event.



NASA Spacesuit User Interface Technologies for Students

Presented by SIRIUS from Northeastern University



In the picture above, draw what you think you would want to see on your augmented reality HUD if you were an Artemis astronaut!

Glossary:

Artemis – Next series of missions to the Moon to establish the Lunar Gateway space station and allow humanity to explore the surface again.

HUD – Heads Up Display, visual elements on a helmet display so a wearer can see information over their view of the world.

Augmented Reality (AR) – Technology that puts digital interactive information over the real world.

Figure A4: Example coloring activity for the Museum of Science Computer Science Weekend.
Coloring image from *Moon Landscape Colouring* (n.d.).

Appendix References

Moon Landscape Colouring. (n.d.). Twinkl.co.uk; Twinkl Educational Publishing. Retrieved Oct. 25, 2024, from <https://www.twinkl.com/resource/moon-landscape-colouring-t-tp-2667096>